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NMR Spectroscopy in an Advanced Inorganic Lab: Structural Analysis of Diamagnetic and Paramagnetic Ni(II) Schiff Base Complexes

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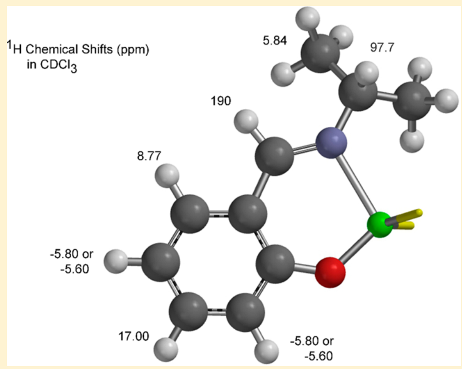
Supporting Information

ABSTRACT: In a two-step synthesis starting with $Ni(CH_3COO)_2{\cdot}4H_2O$ and salicylaldehyde, students prepare *N*-ethyl and *N*-isopropyl Ni(II) Schiff base (imine) complexes. After isolation and recrystallization, the products are characterized by room-temperature magnetic susceptibility measurements, FTIR spectroscopy, and ^{1}H, $^{13}C\{^{1}H\}$, and ^{13}C DEPT-135 or ^{13}C APT NMR spectroscopy. The results are analyzed by considering the steric requirements of the ethyl and isopropyl groups and the subsequent effect on the preferred geometries, either the diamagnetic square-planar geometry, as expected for a Ni(II) d^8 complex, or the paramagnetic tetrahedral geometry. Students are challenged to explain why the spectra for these similar complexes are so different and what this difference reveals about steric requirements in determining geometry.



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INTRODUCTION

Within the undergraduate curriculum, the introduction to NMR spectroscopy typically focuses on ^{1}H and ^{13}C nuclei for the analysis of organic molecules, often those synthesized by students in a traditional organic chemistry course. Even in instrumental analysis and physical chemistry courses, the focus remains on NMR spectroscopy for the determination of diamagnetic organic structures. While NMR spectroscopy of paramagnetic organic molecules is restricted because of inherent instability or high reactivity, transition metal complexes are ideal candidates for the introduction of this topic.

Recent reports in this *Journal* involving NMR spectroscopy of paramagnetic complexes include a lab experiment describing the role of oxidation states of a Co complex to show differences in ^{1}H NMR spectra for diamagnetic and paramagnetic complexes;[1] a synthesis experiment involving various first-row transition metals complexed to the same ligand demonstrating NMR and EPR spectroscopies on diamagnetic and paramagnetic complexes;[2] a computational lab in which $(\eta^5\text{-}C_5H_5)_2M$ (M = V, Fe, Ni) complexes showcase changes in ^{1}H and ^{13}C chemical shifts as a function of the ground-state configuration on M;[3] and a report describing how Schiff base complexes of various transition metals might be used to teach stoichiometry.[4] Schiff base complexes are easily prepared and represent a convenient means to introduce students to a research area with applications from antimicrobial activity to polymerization catalysts.[4,5]

While NMR spectroscopy on paramagnetic species is restricted for most organic free radicals, the technique can be employed with metalloproteins[6] and simple transition metal complexes.[1−4] The spectrum offers a convenient method to determine the magnetic susceptibility using the Evans method,[7] and it is possible to compute spectra for paramagnetic molecules.[3,8]

The difficulties of obtaining NMR spectra for paramagnetic substances are well-established. In the presence of an unpaired electron, the relaxation rate for the excited nucleus is so rapid that the resonance lines are broadened and often indistinguishable from the baseline. Even when detectable, the resonances are unusually broad, and the chemical shifts do not look familiar. Early attempts to describe the origin of chemical shifts and shielding relied upon Ramsey's equation, which includes a diamagnetic contribution dominated by electron density around the observed nucleus and the paramagnetic contribution, which is related to several factors, none of which are clearly diagnostic of geometry.[9] Current descriptions of chemical shifts for paramagnetic molecules divide shift contributions into two components: the contact shift, σ_c, associated with the density of the unpaired electron on the observed nucleus, and a pseudocontact term, σ_p, associated with a through-space interaction between the unpaired electron and the observed nucleus.[8b] A recent paper appeared

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in this *Journal* in which DFT calculations were used to describe ^{1}H and ^{13}C chemical shift changes for a series of structurally known complexes, M(II)(η^5-C_5H_5)$_2$ (M = V, Fe, Ni).[3] However, chemical shifts in a paramagnetic complex do not readily correlate with structural features in an unknown molecule.

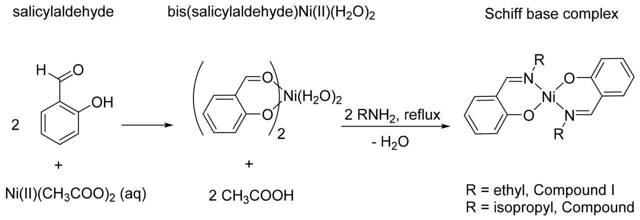
Original studies of Schiff base complexes in the late 1950s and early 1960s employed UV−vis absorption spectroscopy, ^{1}H NMR spectroscopy, and magnetic susceptibility. These discussions represent an important period in the discovery of the fundamental behavior of Ni(II) complexes, a period that included the statement "it should be remembered that the existence of tetrahedral Ni(II) complexes has not been proven and at present it is being more and more regarded as doubtful."[10] Single-crystal X-ray diffraction was used later by the same group to show that the Ni(II) *N*-isopropyl imine complex is in fact tetrahedral.[11] The papers chronicle a critical analysis of data that would be a useful, albeit lengthy, read for today's students. The introduction of optically active R groups within the Schiff base ligand eventually led to the first detection of diastereomers by NMR spectroscopy.[12] The solution behavior of tetracoordinate Ni(II) is complex, and variations in magnetic susceptibility depend on the Lewis basicity of the solvent, temperature, nature of the R group, physical state, and even concentration.[10,13,14] The complexes in this experiment exhibit relatively simple behaviors explained by the steric demands of the R groups alone.

The use of closely related molecules is a pedagogically useful means of guiding student inquiry to identify structure−activity relationships. The present contribution describes the use of Ni(II) Schiff base complexes to demonstrate how steric effects may determine the geometry and spin multiplicity of Ni(II) and the effect on ^{1}H NMR spectra. The two Ni(II) d^8 Schiff base (imine) complexes, bis[2-[(ethylimino-*κN*)methyl]-phenolato-*κO*]nickel(II) (compound I) [13987-25-2]) and bis[2-[((prop-2-yl)imino-*κN*)methyl]phenolato-*κO*]nickel(II) (compound II) [35968-67-3]), which are diamagnetic and paramagnetic, respectively, are easily prepared. The role of steric effects in S_N1 and S_N2 reactions is introduced to students in a typical organic chemistry course, and here the concept is extended to effects on geometry, crystal field theory (CFT) splitting diagrams, and magnetism for the two Ni(II) complexes. A comparison between similar complexes where steric differences determine the geometry using magnetic susceptibility measurements and NMR spectra has not been previously presented as an undergraduate lab.

This lab experiment offers students an opportunity to see ^{1}H and ^{13}C NMR spectra of a paramagnetic complex and to explain the differing Ni(II) geometries. The use of ^{13}C DEPT-135 (distortionless enhanced polarization transfer) or alternatively ^{13}C APT (attached proton test) on the diamagnetic complex (R = ethyl) provides hands-on practice for students in using advanced techniques for assigning carbon resonances.[15]

In the advanced inorganic lab course, this pair of Ni(II) Schiff base complexes is used to illustrate how magnetic susceptibility is related to geometry. In two short lab sessions (2 hours per session) with additional time devoted to characterization, the two complexes are synthesized, purified, and characterized by students; the preparation from a primary amine, either ethyl- or isopropylamine, in a reaction with the hydrated bis(salicylaldehydato)Ni(II) complex Ni(II)-(sal)$_2$(H_2O)$_2$ is shown in Scheme 1. The Schiff base complexes are stable, and samples from previous years may be used if the

Scheme 1. Two-Step Reaction Scheme for the Preparation of Ni(II) Schiff Base Complexes



desire is to focus on a quick introduction to NMR spectroscopy and magnetic susceptibility. Steric factors in the formation of tetrahedral and square-planar complexes for d^8 Ni(II) are explored by students. The focus is the correlation of solid-state magnetic susceptibility with the structure of Ni(II) d^8 complexes and an introduction to NMR spectroscopy beyond that of standard diamagnetic organic molecules. This is a discovery lab experiment in which the starting point for analysis is based on predictions for a Ni(II) d^8 complex; students anticipate two square-planar diamagnetic complexes. Differing steric requirements for the R groups result in the adoption of a square-planar geometry for compound I, the diamagnetic ethyl derivative, and a tetrahedral geometry for compound II, the paramagnetic isopropyl derivative. CFT is used to correlate the geometry with themagnetic susceptibility and NMR behavior for the two complexes.

■ THE EXPERIMENT

This experiment has been part of the advanced inorganic lab course for 35 years, with over 100 students undertaking the lab; it has been modified with changes in technology, particularly the NMR component, and adapted to emphasize the pedagogy found to be the most successful. Although the NMR spectra (Figures S3−S6) were collected on a high-field (400 MHz, 9.4 T) spectrometer, this is not a requirement. Early versions of this lab employed a Varian EM-360 60 MHz continuous-wave spectrometer for ^{1}H NMR spectra only; the increasingly familiar benchtop NMR spectrometers would offer sufficient sensitivity and field dispersion. The Supporting Information (SI) provides detailed student instructions for the lab (pp S2−S4) as well as instructor's notes (pp S7−S10). The experiment is used in the middle of the term after students have been exposed to the background information on coordination chemistry and practiced measurements and calculations for magnetic susceptibility.

Synthesis

The synthesis from readily available reagents is detailed in the SI. It was adopted from the original publications[10,13] with minor changes to address the reaction scale and a safety issue involving the reflux of the primary amine and the bis-(salicylaldehyde) Ni(II) complex. Sample purity is particularly important for susceptibility and NMR measurements, and samples may have to be crystallized more than once. Recrystallization is rapid and routinely affords single crystals 2−3 mm on an edge. Two useful references are provided to students for the identification of trace impurities frequently observed in NMR spectra.[14]

Magnetic Susceptibility

The determination of room-temperature magnetic susceptibility is straightforward and a familiar feature of inorganic experiments. Prior to this lab experiment, magnetic suscept-

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ibility has been used by students to characterize the magnetic state of transition metal complexes (e.g., $[Cu(OAc)_2 \cdot H_2O]_2$); students are already versed in the calculations. A magnetic susceptibility balance previously calibrated with $Hg[Co(SCN)_4]$ is used to determine the susceptibilities of the crystalline products. For magnetic susceptibility calculations, ref 16 is useful for its presentation of consistent values for Pascal's constants, and it includes an example calculation of magnetic moment starting with typical laboratory data. Pascal's constants are used to correct the observed molar susceptibility for diamagnetic contributions.[16] The results are compared to the magnetic susceptibilities of square-planar ($\mu_s = 0$) and tetrahedral ($\mu_s = 2.83\mu_B$) d^8 complexes. Early research on a series of similar Schiff base complexes indicated an equilibrium in solution between the diamagnetic square planar and paramagnetic tetrahedral geometries;[12a,17] there is no evidence for a mixture in the solid state for the two complexes used here.

FT-IR Spectroscopy

Infrared spectra of the crystalline products were collected on an FTIR spectrometer using the attenuated total reflectance (ATR) assembly. Students have extensive experience in assigning IR absorbance peaks and need no special instruction for this step. A useful IR spectroscopy reference for both theory and applications in inorganic compounds is that of Nakamoto.[18]

NMR Spectroscopy

The emphasis in this experiment is the comparison of NMR spectra for two presumably similar complexes and the eventual recognition of how the geometry is correlated with the spectra. To encourage collaboration and discussion, data are collected by student groups for the two imine complexes and shared, and the results are discussed together. The ^1H, ^{13}C, and ^{13}C DEPT-135 or ^{13}C APT spectra on both Schiff base complexes ($CDCl_3$) are collected by the students. Images of student spectra complete with impurities are shown in Figures S3–S6. Except for the expanded spectral range needed for the ^1H NMR spectrum for compound II, the NMR data collection uses default conditions for each experiment.

■ HAZARDS AND SAFETY CONSIDERATIONS

Although none of the reagents used present particular hazards, students must wear nitrile gloves and goggles and perform all synthesis operations in the hood. A useful, searchable online site is available for coded hazard information.[19] Salicylaldehyde [90-02-8], nickel(II) acetate tetrahydrate [6018-89-9], ethanol (95%) [64-17-5], ethylamine (70% in water) [75-04-7], isopropyl amine [75-31-0], $CHCl_3$ [67-66-3], cyclohexane [110-82-7], and $CDCl_3$ [865-49-6] were all used as received. The procedure uses no open flames or high temperatures (>76°), which might be a hazard for the flammable reagents salicylaldehyde, isopropylamine, hexane, and ethanol. The 70% aqueous solution of ethylamine is not flammable. The nickel(II) acetate is a dermatitis hazard and should be handled only while using standard protective gear. No precautionary information appears in the literature for any of the three nickel complexes used here, but exposure to Ni salts has been known to trigger sensitization to dermatitis; students should be required to wear disposable nitrile gloves along with goggles when handling such materials. Filtrates containing Ni(II) should be collected in a labeled waste container and disposed of properly. Recrystallization of the two products uses minimal amounts (<2 mL each) of hexane and $CHCl_3$ and should only be performed in a hood. Hazards associated with $CHCl_3$ (potential carcinogen) and hexane (potential neurotoxin) are associated with prolonged exposure. However, it is possible to avoid either of these two solvents, and alternative crystallization methods affording smaller but acceptable crystals are described in the instructor's notes (p S8 in the SI).

The addition of a splash arrestor (e.g., rotary evaporator, antisplash) to the top of the reflux column to deal with bumping that may occur during reflux (synthesis, step 2) is recommended (Figure S1). Even with boiling stones, the solution shows a tendency to bump violently during reflux and eject material from the top of the reflux column. If a splash arrestor is not available, the glassware arrangement depicted in Figure S2 is an alternative.

■ RESULTS AND DISCUSSION

The reactions provided high-yield routes (>60% in the literature and typically 40–60% for students after obtaining a second crop of crystals) to the two bis(imine) complexes, and the complexes were readily recrystallized by one of several methods described in the instructor's notes (p S8 in the SI). Room-temperature measurements of magnetic susceptibility yielded student values typically within 75% of the expected values. Common experimental errors included incompletely dried samples or large air spaces in the sample tube.

The ^1H NMR spectrum for compound I, the ethyl imine, was readily interpreted (see Table S1). In contrast, the ^1H NMR spectrum for compound II, the paramagnetic isopropyl imine complex, was fundamentally different from what students were expecting.

Surprisingly, all seven expected ^1H resonances for compound II were observable, although students were challenged to find two of them. Students saw broad, featureless resonances (Figure S5a,b) over a chemical shift range dramatically different from that of familiar organic molecules. The resonances for compound II in $CDCl_3$ ranged from −6 to +190 ppm relative to tetramethylsilane. Integration of the peaks did not allow for unambiguous assignments, although the peak areas for proton resonances other than the H–C=N and N–CH moieties were reasonable. The latter two groups are closely associated with the Ni(II) ion. None of the familiar coupling information was observed. However, the absence of the familiar features in the ^1H NMR spectrum was a clear indication that something was fundamentally different between compound I, the ethyl imine, and compound II, the isopropyl imine, and the spectrum did correlate with the solid-state magnetic susceptibility. Discussion focused on why this spectrum was so different from that of the ethyl complex. What was it about this complex that resulted in such an unusual ^1H NMR spectrum? Why should the presence of unpaired electrons, presumably associated with the Ni(II) center, affect the resonance behavior of the hydrogen nuclei?

The ^{13}C and DEPT-135 or APT NMR spectra for compound I allowed assignment of most of the carbon resonances, although not without some ambiguity; Table S2 shows the assignments. The ^{13}C NMR spectrum for compound I was readily interpreted. The two quaternary carbons of the aromatic ring were not observed; additionally, the resonance for a third aromatic carbon, a CH moiety, was not observed. We had no explanation for the latter, although coincidental overlap, rare in ^{13}C spectra, was a possibility. Quaternary carbons within aromatic rings are frequently nonobservable

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low-intensity peaks. They lack the nuclear Overhauser effect (NOE) enhancement associated with proton decoupling.[20]

Description of the Student Population

The students were typically chemistry/biochemistry majors taking this lab course along with a modern inorganic chemistry course. Most had completed both semesters of physical chemistry. Under the more flexible American Chemical Society Committee on Professional Training curriculum, a student taking this course may not have completed the instrumental chemistry course. NMR spectroscopy was previously introduced to students in organic chemistry; lab reports, including literature searches, had been prepared in earlier courses. In a second-year course, coordination chemistry and crystal field theory for the octahedral, tetrahedral, and square-planar geometries had been briefly introduced to students, and in the lecture course accompanying this lab course the details of coordination complexes, including their magnetic behavior, had been presented earlier. Additionally, measurements and calculations of the magnetic susceptibility of the $[Cu(CH_3COO)_2(H_2O)]_2$ complex had been practiced by students. Work was performed by students in groups of two or three for the lab experiment and groups shared information; collaborative data analysis was expected and strongly encouraged.

Guided Discussion

Data collection was relatively straightforward, although the analysis was less so. Data and ideas were shared by students about why the two Schiff base complexes were so different. From their experiences in organic chemistry, they could identify the steric demands of ethyl and isopropyl groups and see how this led to different geometries and crystal field splitting diagrams and ultimately to different spin states for the Ni(II) ion. Still, they needed guiding questions to explain why compound II with two unpaired electrons exhibited such an unfamiliar spectrum. In exploring the ^1H NMR spectrum for compound II, it was unlikely that students would readily locate the proton resonances for the HC═N or the N−CH within the isopropyl group; these were found at 190 and 97.7 ppm, respectively! However, guided discussions were extremely useful in demonstrating how the steric demands led to different ^1H and ^{13}C NMR spectra for the two complexes. Student response to the novel features of this lab was consistently very good. The Supporting Information (pp S5−S6) includes a list of potential questions that may be provided to students on an as-needed basis to guide their discussion.

Student Reports and Goals

Publication-ready reports were prepared in the format of ACS publications using *Inorganic Chemistry* as a guideline. Reports included a detailed discussion of safety information, including references to MSDS information and any unusual safety procedures such as the addition of the splash arrestor on the reflux column. Reports included references to the primary literature accessible through online searches using, e.g., SciFinder. It was important that detailed comments from the instructor were provided on the work. Students at the end of the course have indicated that this feature of the lab, in which they received a critical reading of their manuscript, including early drafts, and detailed feedback in writing, was particularly useful.

The value of the lab report, separate from the preparation of a well-written report, was accessed by how the students achieved the following goals:

1. the synthesis and purification of two Ni(II) complexes, both of which are predicted to be square planar and diamagnetic (5);
2. the measurement and correlation of magnetic susceptibility, μ, with the appropriate CFT splitting diagram (4−5);
3. the recognition that μ for compound I, the ethyl derivative, is consistent with square-planar geometry (4−5);
4. the recognition that μ for compound II, the isopropyl derivative, is consistent with tetrahedral geometry (4−5);
5. the proposition that steric differences between the ethyl and isopropyl derivatives determine the observed geometries (5);
6. the recognition that ligand steric requirements determine the CFT splitting diagram and the observable magnetic susceptibility (4−5);
7. the observation that diamagnetic and paramagnetic susceptibilities are reflected in the differences between the NMR spectra for the two compounds (3−4);
8. assignment of proton and carbon resonances for spectra where possible (3−4);
9. the use of a literature search to propose how the presence of unpaired electrons affects the chemical shifts and line shapes of NMR resonances (3).

The number in parentheses (scaled 1−5, with 5 representing successful achievement) after each goal represents an assessment of how successfully the goal was achieved over the decades. The syntheses always yielded the intended products, and recrystallization attempts routinely provided high-quality crystals in reasonable yields. The connection between the experimental measurement of magnetic susceptibility and the correct CFT diagram was readily made by students. It was the experimental observation that drove the choice, not their preconception. The steric influences in the reaction represented a familiar concept from student experiences in a typical organic course. In goals 5 and 6 the expectation was that several concepts must come together: a connection must be made between steric requirements, complex geometry, magnetic susceptibility, and the CFT diagram. Steric requirements drove the geometry that determined the observed magnetic behavior, which was explained by the splitting diagram. The NMR spectra for compound I were readily assigned on the basis of what students learned in an organic chemistry course. The spectra for compound II were problematic and represented an unfamiliar paradigm for NMR spectroscopy. Students were presented with an observation in which a memorized preconception concerning chemical shifts and line shapes was clearly not evident. Goal 7 was a step toward how to interpret actual research observations where multiple pieces of information were brought together into a consistent description in a lab report on the structure of two Ni(II) Schiff base complexes. Historically, this was an area in which students omitted discussion items; the student focus was often on the conclusion alone. Overall, there must be a recognition that transition metal complexes of the same metal exhibit different geometries based on steric demands of two similar ligands and that these steric differences resulted in observable magnetic susceptibility and NMR spectral differences. A successful lab report was one in which these individual goals were successfully addressed. The guided inquiry

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questions (pp S5−S6 in the SI) were designed to help students reach this conclusion. It was comforting when one student, even if only one, said afterward, "now NMR makes sense to me."

■ CONCLUSION

This discovery lab was a pedagogically useful way to introduce a variety of NMR techniques for the characterization of inorganic complexes prepared by the students. The synthesis of closely related complexes in which steric requirements of the ligands yield different molecular geometries provided students an opportunity to correlate steric demands, complex geometry, magnetic susceptibility, and NMR behavior. The ^{1}H and ^{13}C NMR spectra provided students with practice in interpreting the spectra for a diamagnetic complex and served as an introduction to the NMR spectral properties of the closely related paramagnetic complex. All three NMR spectral types (^{1}H, ^{13}C, and ^{13}C DEPT or APT) were useful techniques for characterizing transition metal complexes. The ^{1}H NMR spectra for both complexes revealed that the solid-state and solution behaviors were magnetically consistent and confirmed the tetrahedral and square-planar geometries. The ^{1}H NMR spectrum for the *N*-isopropyl complex was particularly useful as a teaching tool for its contrast to a familiar, "normal" ^{1}H NMR spectrum.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available on the ACS Publications website at DOI: 10.1021/acs.jchemed.8b00399.

> Lab procedure (three-page handout for students, pp S2−S4); guided inquiry questions (two-page optional handout for students, pp S5−S6); instructor's notes (four-page guideline, pp S7−S10); example NMR spectra (seven annotated figures of ^{1}H and ^{13}C NMR spectra, pp S11−S18) (PDF, DOCX)

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Notes

The author declares no competing financial interest.

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